

In this exercise, you'll enhance your multi-modal chatbot by adding voice input and output capabilities. Users will be able to speak to your chatbot instead of typing, and receive spoken responses, creating a truly conversational AI experience.

What You'll Learn

By completing this exercise, you will:

- **Implement** speech-to-text functionality for voice input
- **Add** text-to-speech capabilities for voice output
- **Create** a hands-free conversational experience
- **Experience** browser-based voice API integration with Kiro

Step 1: Add Voice Input

Start by adding the ability for users to speak to your chatbot:

Now I would like to add the ability to talk to the chatbot, instead of typing. Please implement voice input functionality so users can speak their messages.



⚠️ Browser Compatibility

Voice features may only be available in certain browsers like Chrome. Kiro will implement the best available solution for cross-browser compatibility.

Step 2: Add Voice Output

Once voice input is working, enhance the experience with voice responses:

Now I would like to add the ability for the bot to talk back in voice as well. Please implement text-to-speech so the chatbot can speak its responses.



What Kiro Will Update

Your voice enhancement will include:

- **Voice Input:** Speech-to-text functionality using browser APIs
- **Voice Output:** Text-to-speech for chatbot responses
- **UI Controls:** Voice recording and playback controls
- **Browser Integration:** Web Speech API implementation
- **Fallback Options:** Graceful degradation for unsupported browsers

✅ Complete Voice Experience

Your chatbot will now support both listening to users and speaking responses, creating a natural conversational interface.

Step 3: Test Voice Functionality

Please test both voice input and output features. Make sure you can speak to the chatbot and hear responses back.

Expected Behavior

Your enhanced chatbot should demonstrate:

- ▶ **Voice Input Example**
- ▶ **Voice Output Example**

Step 4: Update Documentation

Since you've built an impressive multi-modal AI chatbot, update your project documentation:

Can you generate or update the readme file with the details of what we have done? Include all the features we've implemented throughout the workshop.



Success Criteria

After completing this exercise, you should have:

- ☒ **Voice Input:** Users can speak messages to the chatbot
- ☒ **Voice Output:** Chatbot responds with spoken audio
- ☒ **Browser Integration:** Proper use of Web Speech APIs
- ☒ **UI Controls:** Intuitive voice recording and playback interface
- ☒ **Updated Documentation:** Comprehensive README reflecting all features

✅ Voice-Enabled AI Complete

Your chatbot now offers a complete hands-free conversational experience with voice input, text/image generation, and voice output!

Troubleshooting

Common issues you may encounter with voice features:

- ▶ **Browser Compatibility Issues**
- ▶ **Microphone Access Denied**
- ▶ **Voice Recognition Accuracy**
- ▶ **Audio Playback Issues**

Architecture Overview

- ▶ **View Complete Voice-Enabled Architecture**

This exercise completes your transformation from a simple text chatbot to a sophisticated multi-modal AI assistant with voice, text, and image capabilities.

Key Takeaways

This exercise demonstrated advanced browser integration and voice technology concepts:

Voice Technology

User Experience

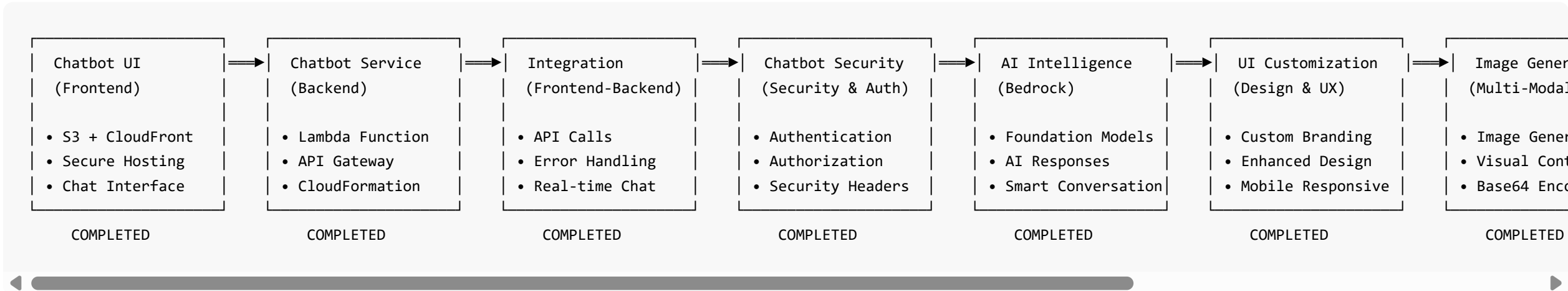
Technical Integration

- **Speech Recognition:** Implemented browser-based speech-to-text conversion
- **Text-to-Speech:** Added natural voice output for chatbot responses
- **Browser APIs:** Utilized Web Speech API for voice functionality
- **Real-time Processing:** Handled live audio input and output streams
- **Cross-Browser Support:** Implemented compatibility across different browsers

Next Steps

Your chatbot now has comprehensive voice capabilities, making it a truly hands-free AI assistant. The next exercise will add user authentication to make your application enterprise-ready with secure user management.

Workshop Progress



[Let's Implement User Registration and Authentication](#)

Previous

Next